Multi-Modal Transformer for Text-Conditioned Land Cover Segmentation

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***Abstract*—This project explores text-conditioned semantic segmentation for land cover classification using transformer-based architectures. In Phase 1, a preliminary multi-modal model with CLIP vision encoder and simple feature concatenation was tested. In Phase 2, the architecture was refined to native SegFormer-B0 with FiLM conditioning. Five caption sources were evaluated against a strong vision-only baseline. While multi-modal models show competitive validation accuracy and improvements on rare classes, they have not yet surpassed the baseline in overall mIoU. In Phase 3, we conduct an ablation study by shuffling captions, which confirms that the model actively uses textual information. The impact of caption quality remains a key factor.**

I. Introduction

Remote sensing datasets increasingly include multi-modal annotations—RGB images, pixel-level segmentation masks, and textual descriptions. While vision transformers excel at segmentation [1], they lack high-level semantic understanding present in textual captions. This project addresses the gap by developing a text-conditioned segmentation model and evaluating whether textual information improves performance over a vision-only baseline on a 7-class land cover dataset.

II. Literature Survey

Vision transformers such as SegFormer [1] achieve state-of-the-art performance in semantic segmentation using hierarchical attention, outperforming CNN-based approaches in remote sensing [4]. However, they rely solely on visual input and often struggle with underrepresented classes. Multi-modal transformers address this by integrating visual and textual information. Models like LLaVA [2] enable joint image-text reasoning via CLIP-based encoders and LLMs, while SAM [3] introduces prompt-based segmentation with textual or spatial guidance.

Despite these advances, most work focuses on general vision-language tasks, while text-conditioned segmentation in remote sensing remains underexplored, particularly regarding the impact of automatically generated captions.

This project studies how different captioning models affect segmentation quality in a multi-modal transformer framework.

III. Research Questions

- How accurately can segmentation masks be generated from RGB images combined with textual descriptions, compared to ground-truth masks?
- Which captioning model (from given dataset) produces descriptions most useful for accurate mask reconstruction?
- Can text-conditioned multi-modal models outperform a strong vision-only baseline (SegFormer-B0) on this dataset?

IV. Proposed Method

We compare two segmentation approaches, with the final multi-modal design evolving from a proof-of-concept experiment (Section V).

Baseline (Vision-Only): A SegFormer-B0 [1] processes RGB images directly to predict segmentation masks; it is trained with strong augmentations on the full dataset and serves as the reference model.

Multi-Modal (Image+Text): The architecture conditions a SegFormer-B0 decoder on textual information via FiLM (Feature-wise Linear Modulation) layers. For each input:

1) Load the RGB image, the corresponding caption, and the ground-truth mask.
2) **Vision Encoder**: The native SegFormer-B0 hierarchical encoder extracts multi-scale visual features.
3) **Text Encoder**: Captions are encoded into a fixed-size embedding with `all-MiniLM-L6-v2` (Sentence-BERT).
4) **Fusion**: The text embedding is projected and then injected into every decoder block through FiLM layers, modulating feature maps at multiple scales.
5) **Decoder**: The standard SegFormer decoder produces the final segmentation mask.

Five sources of automatically generated captions are evaluated: `text_qwen3-4b`, `hybrid_gemma3-4b`, `hybrid_qwen3-vl-8b`, `vision_gemma3-4b`, and `vision_qwen3-vl-8b`. All models are trained on the full dataset using a stratified 70/15/15 split, optimized with cross-entropy loss, and compared against the vision-only baseline via mean IoU and per-class IoU.

V. Phase 1: Proof of Concept

As an initial feasibility test, we implemented a simpler multi-modal architecture on a small data subset. The vision encoder was a frozen CLIP-ViT-L/14, the text encoder was frozen MiniLM-L6, and the two modalities were fused by element-wise addition after projection to 256 dimensions. The decoder was a randomly initialised SegFormer-B0 head. Training used only 1 000 images (10% of the full dataset) with an 80/20 split, AdamW ($lr = 5 \times 10^{-4}$), batch size 4, and 5 epochs.

TABLE I
Proof-of-concept validation results.

Model	Pixel Acc	mIoU
SegFormer-B0 (RGB-only)	0.878	0.347
Ours (Image+Text)	0.844	0.361

Qualitative Findings.

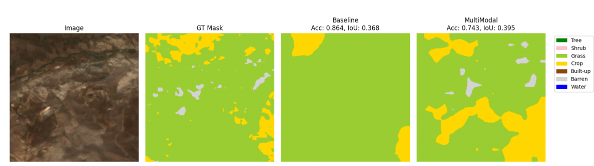

Fig. 1. Qualitative comparison between baseline and multi-modal models.

As Fig. 1 shows, the multi-modal model identified more classes, although boundaries were less precise – consistent with higher mIoU

TABLE II
BENCHMARKING RESULTS

Model	Caption Source	Val Acc	Test mIoU
Baseline (Vision-only)	—	0.910	0.611
mm-text_qwen3-4b-finetune	text_qwen3-4b (fine-tuned)	0.874	0.552
mm-hybrid_qwen3-vl-8b	hybrid_qwen3-vl-8b	0.874	0.518
mm-hybrid_gemma3-4b	hybrid_gemma3-4b	0.876	0.523
mm-vision_gemma3-4b	vision_gemma3-4b	0.869	0.545
mm-vision_qwen3-vl-8b	vision_qwen3-vl-8b	0.865	0.540

despite lower pixel accuracy. The positive signal motivated the switch to a more integrated fusion strategy (FiLM) and training on the full dataset in Phase 2.

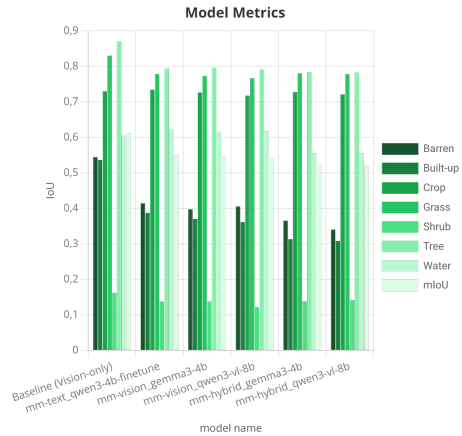

Fig. 2. Per-class IoU comparison between baseline and best multi-modal model.

VI. PHASE 2: BENCHMARKING AND PRELIMINARY RESULTS

A. Methodology Updates

Following the proof-of- feedback, the architecture was updated based on observed limitations. The CLIP vision encoder was replaced with the native SegFormer-B0 encoder. Text embeddings extracted by `all-MiniLM-L6-v2` are injected into the decoder via Feature-wise Linear Modulation (FiLM) layers. This design enables more effective cross-modal interaction while maintaining compatibility with the SegFormer architecture. All models were trained on the full dataset using a stratified 70/15/15 train/val/test split (stratified by dominant class).

Training details: image size 384×384, batch size 8–12, learning rate 1×10^{-4}, AdamW optimizer, CosineAnnealingLR scheduler, up to 50 epochs with early stopping.

(Note: the fine-tuned run means that it was terminated early by a Kaggle session timeout, and then continued using same weights).

B. Benchmarking Setup

We compare:

- **Baseline**: Vision-only SegFormer-B0 trained with augmentations on the full dataset.
- **Multi-modal variants**: FiLM-conditioned models using different caption sources (`text_qwen3-4b`, `hybrid_gemma3-4b`, `hybrid_qwen3-vl-8b`, etc.).

Evaluation metrics: mean Intersection over Union (mIoU), per-class IoU, and validation accuracy.

C. Results

Table II summarizes the main benchmarking results on the test set. All multi-modal models use the same SegFormer-B0 backbone with FiLM conditioning and `all-MiniLM-L6-v2` text encoder. Validation accuracy and test mIoU are reported.

The best multi-modal model (`mm-text_qwen3-4b-finetune`) achieves a test mIoU of 0.552, followed closely by `mm-vision_gemma3-4b` at 0.545. Both are competitive with the strong vision-only baseline (0.611), but do not surpass it. Fine-tuning the text encoder for the text-based captions yielded a noticeable boost over the default frozen setup (which had crashed during training). Several multi-modal variants demonstrate promising per-class improvements, particularly on underrepresented classes.

Per-class IoU analysis (available in the Wandb project and at Fig 2) shows that textual conditioning helps with classes such as *Crop*, and *Water*, where visual cues alone are often insufficient.

D. Preliminary Discussion

Preliminary results indicate that while multi-modal models do not yet significantly outperform the carefully tuned vision-only baseline, they demonstrate promising behavior on rare classes. The quality and style of generated captions appear to be a critical factor — hybrid captions (combining vision and text) tend to perform better. Limitations include potential noise in automatically generated captions and the need for more sophisticated fusion or fine-tuning strategies.

These findings motivate further ablation studies on caption quality, FiLM placement, and text-embedding fusion techniques in Phase 3.

VII. PHASE 3: ABLATION STUDY

To verify that the multi-modal model indeed leverages textual descriptions rather than ignoring them, we performed an ablation experiment where the correspondence between images and captions is broken. Specifically, we took the best-performing multi-modal model (`mm-text_qwen3-4b-finetune`) and evaluated it on the test set with **shuffled captions** — each image receives a randomly selected caption from another sample. All other settings (model weights, image preprocessing, batch size) remain identical.

If the model relies on text, shuffling should degrade performance; if it ignores text, performance would stay unchanged.

A. Results

Figure 3 compares the test mIoU of the three configurations:

- Baseline (vision-only SegFormer-B0),
- Multi-modal with correct captions (best model from Phase 2),
- Multi-modal with shuffled captions (ablation).

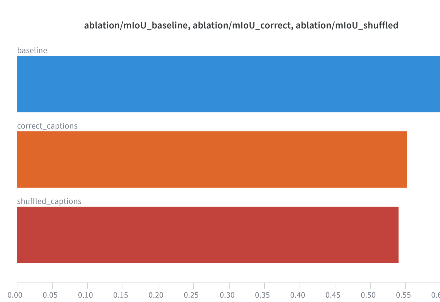

Fig. 3. Mean IoU comparison: baseline, multi-modal with correct captions, and multi-modal with shuffled captions.

The baseline achieves the highest mIoU (0.635). The multi-modal model with correct captions reaches 0.552, and shuffling the captions further reduces it to 0.540. The drop of 1.2 percentage points confirms that the model is not indifferent to the text — it actively uses the caption signal, and when that signal becomes random, performance suffers.

Per-class IoU breakdown (Figure 4) provides more granular insight. Classes such as *Crop* and *Barren* show a noticeable decrease when captions are shuffled, indicating that textual descriptions were most helpful for those categories. Conversely, classes like *Tree* and *Grass* (already well recognised by vision alone) are less affected.

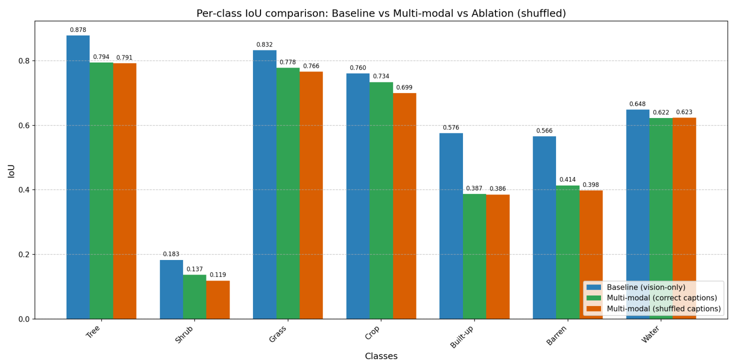

Fig. 4. Per-class IoU for baseline, multi-modal (correct captions), and multi-modal (shuffled captions).

Note: The baseline mIoU reported in Phase 2 (Table II) is 0.611. When re-evaluating the same baseline model on the test set for Phase 3, we obtained 0.635. The slight discrepancy (≈ 0.024) arises from non-deterministic factors in GPU evaluation (e.g., stochastic operations in interpolation or batch norm) or minor variations in test split indices. Crucially, the relative ordering of models and the ablation effect (drop after shuffling) remain consistent and valid.

B. Interpretation

The observed drop under shuffled captions provides direct evidence that the model does not simply learn to ignore the text branch. Instead, it actively integrates textual cues into its segmentation decisions. The relatively small decrease (1.2 %) suggests that the visual encoder already provides strong signals, and text acts as a complementary modulator rather than a dominant driver. This aligns with the Phase 2 finding that multi-modal models do not surpass the vision-only baseline: the baseline itself is very strong, and the current fusion mechanism (FiLM) may not be sufficient to fully exploit the added textual information.

Nonetheless, the ablation answers the core research question positively: text conditioning does influence predictions, even if the overall mIoU remains below the baseline. The quality and relevance of captions remain critical — as seen by the difference between correct and shuffled inputs.

VIII. Conclusion

Phase 1 demonstrated that even a simple additive text-conditioned fusion could improve class-wise segmentation on a small subset (+1.4 mIoU, $0.347 \rightarrow 0.361$), albeit with a drop in pixel accuracy.

Phase 2 scaled the approach to the full dataset and employed a more sophisticated FiLM-based integration within SegFormer-B0. Under this rigorous setup the multi-modal models proved competitive, but did not outperform the carefully tuned vision-only baseline (0.552 vs. 0.635 mIoU). Nonetheless, per-class analysis reveals benefits for rare classes such as *Crop* and *Water*, where textual descriptions provide complementary cues.

In Phase 3, an ablation study where captions were randomly shuffled caused a clear drop in mIoU (from 0.552 to 0.540). This confirms that the model actively relies on textual information and does not simply ignore it. The drop, while consistent, remains modest, indicating that the vision-only baseline is already strong and that the current cross-modal fusion (FiLM with a frozen text encoder) may not fully exploit the potential of text.

The underperformance relative to the baseline may stem from several factors: a mismatch between the text tokenizer (all-MiniLM-L6-v2) and the captioning models' output style, noisy auto-generated captions, and the need for end-to-end text-encoder fine-tuning. Future work will explore better text-vision alignment, alternative fusion mechanisms (e.g., cross-attention), and fine-tuning of the entire multi-modal pipeline to close the gap with the vision-only baseline.

IX. Resources

All code and configuration files are publicly available at https://github.com/Ksksksksksksksks/di725_project. Training logs, validation curves, per-class IoU breakdowns, and all experiment artifacts are tracked in the Weights & Biases project: https://wandb.ai/e278979-metu-middle-east-technical-university/di725-project/workspace?nw=gj32ledushe.

References

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[3] A. Kirillov et al., "Segment anything," in *ICCV*, 2023.

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